Robust optimization 101

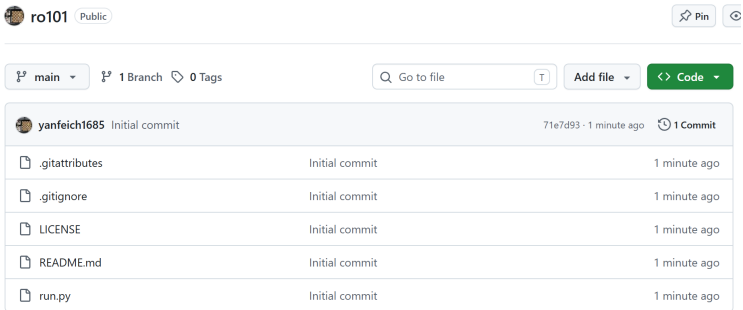
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Github repository

The slides and code for this lecture is available at https://github.com/yanfeich1685/ro101.



Linear programs: Canonical form

A **linear program (LP)** in canonical form is given by

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{c}$ are problem data and $\boldsymbol{x}$ is the vector of decision variables.

Question: Are LPs easy to solve?

Linear programs: Complexity

LPs can be solved in **polynomial time** using algorithms such as the ellipsoid method and the interior point method (IPM).

The famous simplex algorithm generally solves LPs very efficiently in practice, although it runs in **exponential time** in the worst case.

Bottom line: LPs are easy.

Production problem: Deterministic

Our factory needs to produce 100 units of Product A, whose key material is supplied by Suppliers 1–4. Different suppliers quote different prices. The conversion rates also differ.

Table: Material from four suppliers.

Name	Notation	S1	S2	S3	S4
Cost per unit	c_i	3	4	5	6
Conversion rate	a_i	0.4	0.5	0.8	0.9
Order amount	x_i	x_1	x_2	x_3	x_4

The LP is simply

$$\begin{aligned} \min_{\boldsymbol{x}} \ & 3x_1 + 4x_2 + 5x_3 + 6x_4 \\ \text{s.t. } & 0.4x_1 + 0.5x_2 + 0.8x_3 + 0.9x_4 \geq 100 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

Question: Can we solve it without invoking Gurobi?

Production problem: Deterministic

Name	Notation	S1	S2	S3	S4
Cost per unit	c_i	3	4	5	6
Conversion rate	a_i	0.4	0.5	0.8	0.9
Cost per output	c_i/a_i	7.50	8.00	**6.25**	6.67
Order amount	x_i	x_1	x_2	x_3	x_4

The best strategy is to buy the material only from Supplier 3. In the optimal solution, $x_1 = x_2 = x_4 = 0$ and $x_3 = 100/0.8 = 125$. The minimum (optimal) cost is $5 \cdot 125 = 625$.

In this example, all problem data $\boldsymbol{a}, \boldsymbol{c}$ are known with certainty. We call this problem **deterministic**.

Production problem under uncertainty

However, conversion rates can be subject to factors like the weather conditions.

Because we must place our orders before the production process, we do not know $\boldsymbol{a}$ with certainty at the decision moment.

Each possible realization of $\boldsymbol{a}$ is called a **scenario**. Suppose that there are finitely many scenarios. We collect all scenarios in an **uncertainty set** $\mathcal{U} = \{\boldsymbol{a}^1, \ldots, \boldsymbol{a}^P\}$.

We need to make sure that the solution is feasible for all scenarios in $\mathcal{U}$.

Question: What does the production problem become now?

Production problem under uncertainty: Finite scenarios

Write the deterministic LP as

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & \boldsymbol{a}^\top \boldsymbol{x} \geq b \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

Since any feasible solution to the deterministic LP has to remain feasible for every $\boldsymbol{a}^p \in \mathcal{U}$, we **copy** $\boldsymbol{a}^\top \boldsymbol{x} \geq b$ P times. The robust LP is

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & (\boldsymbol{a}^p)^\top \boldsymbol{x} \geq b \quad p = 1, \ldots, P \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

It becomes larger, but **it is still an LP**.

Production problem under uncertainty: Infinite scenarios

However, we can also model each conversion rate as a **continuous variable** within an interval, i.e., $a_i = \overline{a_i} - d_i\xi_i, \xi_i \in [0,1]$, where

- $\overline{a_i}$: maximum conversion rate of Supplier i's material
- $\overline{a_i} - d_i$: minimum conversion rate of Supplier i's material
- d_i: difference between the maximum and minimum conversion rates
- ξ_i: deviation ratio

We do not know ξ_i at the decision moment. The uncertainty set is

$$\mathcal{U}^{\text{box}} = \{\boldsymbol{a} : a_i = \overline{a_i} - d_i\xi_i, \xi_i \in [0,1] \quad i = 1,2,3,4\},$$

and its **primitive** version is

$$\Xi^{\text{box}} = \left\{\boldsymbol{\xi} : \boldsymbol{\xi} \in [0,1]^4\right\},$$

Production problem under box uncertainty

We call $\Xi^{\text{box}} = \{\boldsymbol{\xi} : \boldsymbol{\xi} \in [0,1]^4\}$ a **box uncertainty set**. Mathematically, it is a **hypercube** in $\mathbb{R}^4$.

The robust LP is

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & \overline{\boldsymbol{a}}^\top \boldsymbol{x} - \sum_{i=1}^{4} d_i \xi_i x_i \geq b \quad \forall \boldsymbol{\xi} \in \Xi^{\text{box}} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

Enumeration no longer works because there are **infinitely many scenarios**, i.e, $|\Xi^{\text{box}}| = \infty$. We cannot solve it like this.

Production problem under box uncertainty

Under uncertainty, we have the following data

Name	Notation	S1	S2	S3	S4
Cost per unit	c_i	3	4	5	6
Max conversion rate	$\overline{a_i}$	0.4	0.5	0.8	0.9
Max deviation	d_i	0.1	0.1	0.1	0.05
Min conversion rate	$\overline{a_i} - d_i$	0.3	0.4	0.7	0.85

The robust LP is then

$$\begin{aligned} \min_{x} \; & 3x_1 + 4x_2 + 5x_3 + 6x_4 \\ \text{s.t. } & (0.4 - 0.1\xi_1)x_1 + (0.5 - 0.1\xi_2)x_2 + (0.8 - 0.1\xi_3)x_3 + (0.9 - 0.05\xi_4)x_4 \geq 100 \quad \forall \boldsymbol{\xi} \in [0,1]^4 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

Question: How to get rid of the universal quantifier $\forall$ and solve this?

Production problem under box uncertainty

Take a look at the coefficient of x_i: $\overline{a_i} - d_i\xi_i$ is always positive for all $\xi_i \in [0, 1]$.

In the worst case, each conversion rate takes its minimum value, i.e., $a_i = \overline{a_i} - d_i$. Hence, we solve it just like we did previously.

Name	Notation	S1	S2	S3	S4
Cost per unit	c_i	3	4	5	6
Max conversion rate	$\overline{a_i}$	0.4	0.5	0.8	0.9
Max deviation	d_i	0.1	0.1	0.1	0.05
Min conversion rate	$\overline{a_i} - d_i$	0.3	0.4	0.7	0.85
Max cost per output	$c_i/(\overline{a_i} - d_i)$	10.00	10.00	7.14	**7.06**
Order amount	x_i	x_1	x_2	x_3	x_4

We are all-in on Supplier 4. We buy $100/0.85 = 117.65$ units with a cost of 705.88. In the worst-case scenario where $a_4 = 0.85$, our factory still produces 100 units of Product A.

Is the box a good idea?

Murphy's law: **Anything that can go wrong will go wrong.**

Conservatism is necessary and sometimes nonnegotiable in fields like nuclear reactor or dam construction.

Do we have to assume that all conversion rates take their minimum values **simultaneously**?

If failing to produce 100 units of Product A results in acceptable consequences (delayed consignment, etc.), should we strictly adhere to the box uncertainty set?

Share your thoughts.

Budgeted uncertainty set

Bertsimas and Sim (2004) propose the well-known **budgeted uncertainty set**. In our setting, it is given by

$$\Xi^{\text{budget}} = \left\{ \boldsymbol{\xi} : \boldsymbol{\xi} \in [0,1]^4; \sum_{i=1}^{4} \xi_i \leq \Gamma \right\},$$

where $\Gamma \in \{0, 1, \ldots, 4\}$ is the budget. When

- $\Gamma = 0$, there is no uncertainty, and the problem is **deterministic**
- $\Gamma = 4$, all conversion rates are allowed to reach their minimum values simultaneously: **this is the box uncertainty set**
- $\Gamma = 1, 2, 3$, at most Γ conversion rates are allowed to reach their minimum values simultaneously

The choice of Γ depends on the problem setting and the decision maker's risk attitude.

Production problem under budgeted uncertainty

Omitting the derivation (see Appendix), the production problem under budgeted uncertainty (**robust counterpart**) is given by

$$
\begin{aligned}
\min_{\boldsymbol{x},z,\boldsymbol{q}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{s.t.} \quad & \overline{\boldsymbol{a}}^\top \boldsymbol{x} - \Gamma z - \sum_{i=1}^{4} q_i \geq b \\
& z + q_i \geq d_i x_i \quad i = 1, 2, 3, 4 \\
& \boldsymbol{x}, \boldsymbol{q} \geq \boldsymbol{0} \\
& z \geq 0.
\end{aligned}
$$

It is an LP, so **it is easy**.

Experiments

```
x = model.addMVar(shape=4, lb=0, vtype=GRB.CONTINUOUS) # matrix variable
q = model.addMVar(shape=4, lb=0, vtype=GRB.CONTINUOUS)
z = model.addVar(lb=0, vtype=GRB.CONTINUOUS)
model.setObjective(c @ x, GRB.MINIMIZE) # @: matrix product
model.addConstr(a @ x - Gamma * z - q.sum() >= b)
model.addConstr(q >= d * x - z)
```

Table: Optimal cost and solution under different budgets.

Budget	Optimal cost	x_1	x_2	x_3	x_4
0 (deterministic)	625.00			125.00	
1	**680.00**			**40.00**	**80.00**
2	705.88				117.65
3	705.88				117.65
4 (box)	705.88				117.65

Can we do less math?

We can do robust optimization using less math via

- Box uncertainty
- Simulation
- Vertex enumeration (see Appendix)

Less math via box uncertainty

We simply consider the absolute worst-case scenario.

The solution can be overly conservative.

Sometimes it is the only option.

Less math via simulation

For budgeted uncertainty, we sample scenarios from Ξ^{budget}.

Since $\boldsymbol{\xi}$ is at least as bad as $\boldsymbol{\xi}'$ if $\boldsymbol{\xi} \geq \boldsymbol{\xi}'$ (componentwise), we normalize each scenario such that $\sum_{i=1}^{4} \xi_i = \Gamma$. For example, when $\Gamma = 2$, $(0.2, 0.3, 0.1, 0.4)$ is normalized to $(0.4, 0.6, 0.2, 0.8)$.

Let Ξ^{sampled} be the collection of all sampled scenarios; the robust LP is

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & \overline{\boldsymbol{a}}^\top \boldsymbol{x} - \sum_{i=1}^{4} d_i \xi_i x_i \geq b \quad \forall \boldsymbol{\xi} \in \Xi^{\text{sampled}} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

The optimal solution to this LP is **not** guaranteed to remain feasible for all $\boldsymbol{\xi} \in \Xi^{\text{budget}}$. However, it can be a practical, implementable solution.

Less math via simulation

```
def simulation(Gamma, sce):
    n_sim_list = [10, 50, 100, 500, 1000] # number of scenarios
    for n_sim in n_sim_list:
        model = gp.Model()
        x = model.addMVar(shape=4, lb=0, vtype=GRB.CONTINUOUS)
        model.setObjective(c @ x, GRB.MINIMIZE)
        model.addConstr(a @ x >= b) # maximum conversion rates
        for s in range(n_sim):
            xi = sce[s]
            xi = xi / xi.sum() * Gamma # normalization
            a_new = a - xi * d
            model.addConstr(a_new @ x >= b)
```

More experiments

See Appendix for vertex enumeration.

Table: Optimal cost and solution for $\Gamma = 1$.

Model	Opt cost	x_1	x_2	x_3	x_4	Runtime
Robust counterpart	**680.00**			**40.00**	**80.00**	**0.01**
Simulation (10 scenarios)	674.90			36.51	82.06	0.01
Simulation (50 scenarios)	677.74			28.25	89.41	0.03
Simulation (100 scenarios)	678.15			32.50	85.94	0.04
Simulation (500 scenarios)	678.88			36.94	82.36	0.19
Simulation (1000 scenarios)	**679.04**			**38.05**	**81.46**	**0.42**
Vertex enumeration	680.00			40.00	80.00	0.01

Takeaways

In this lecture, we

- Study a production problem when all problem data are known
- Understand the necessity to consider data uncertainty
- Solve the problem under box uncertainty (the absolute worst-case scenario)
- Study how to adjust conservatism via budgeted uncertainty
- Learn how to do robust optimization in a light-math way

Final message

When doing optimization, ask “are the data known with certainty?”

If the answer is no, consider robust optimization.

References



Bertsimas, D., & Sim, M. (2004). The price of robustness. *Operations Research*, *52*(1), 35–53. https://doi.org/10.1287/opre.1030.0065

Appendix: Derivation of the robust counterpart

Solving the production problem (and a general LP) under budgeted uncertainty requires duality.

The constraint

$$\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - (d_1\xi_1 x_1 + d_2\xi_2 x_2 + d_3\xi_3 x_3 + d_4\xi_4 x_4) \geq b \quad \forall \boldsymbol{\xi} \in \Xi^{\text{budget}},$$

is equivalent to

$$\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - \max_{\boldsymbol{\xi} \in \Xi^{\text{budget}}} \{d_1 x_1 \xi_1 + d_2 x_2 \xi_2 + d_3 x_3 \xi_3 + d_4 x_4 \xi_4\} \geq b,$$

Appendix: Derivation of the robust counterpart

The primal maximization problem

$$\begin{aligned} \max_{\boldsymbol{\xi}} \; & d_1 x_1 \xi_1 + d_2 x_2 \xi_2 + d_3 x_3 \xi_3 + d_4 x_4 \xi_4 \\ \text{s.t. } & \xi_1 + \xi_2 + \xi_3 + \xi_4 \leq \Gamma \qquad (z) \\ & \xi_1, \xi_2, \xi_3, \xi_4 \leq 1 \qquad (q_1, q_2, q_3, q_4) \\ & \xi_1, \xi_2, \xi_3, \xi_4 \geq 0 \end{aligned}$$

has a dual minimization problem

$$\begin{aligned} \min_{z, \boldsymbol{q}} \; & \Gamma z + q_1 + q_2 + q_3 + q_4 \\ \text{s.t. } & z + q_i \geq d_i x_i \quad i = 1, 2, 3, 4 \\ & z, q_1, q_2, q_3, q_4 \geq 0. \end{aligned}$$

Appendix: Derivation of the robust counterpart

The constraint

$$\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - \max_{\boldsymbol{\xi}\in\Xi^{\text{budget}}} \{d_1x_1\xi_1 + d_2x_2\xi_2 + d_3x_3\xi_3 + d_4x_4\xi_4\} \geq b,$$

now becomes the **constraint group**

$$\begin{aligned}
&\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - \min_{z,\boldsymbol{q}} \{\Gamma z + q_1 + q_2 + q_3 + q_4\} \geq b \\
&z + q_i \geq d_i x_i \quad i = 1,2,3,4 \\
&z, q_1, q_2, q_3, q_4 \geq 0.
\end{aligned}$$

The first constraint in the group still needs more work.

Appendix: Derivation of the robust counterpart

Rearranging

$$\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - \min_{z,\boldsymbol{q}} \{\Gamma z + q_1 + q_2 + q_3 + q_4\} \geq b$$

gives

$$\min_{z,\boldsymbol{q}} \{\Gamma z + q_1 + q_2 + q_3 + q_4\} \leq \overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - b.$$

Consider the statement: *If Shanghai Tower is at least as tall as (no shorter than) the shortest skyscraper in Dubai, then Shanghai Tower is at least as tall as (no shorter than) at least one skyscraper in Dubai.*

Table: Shanghai Tower vs. some skyscrapers ($\geq$ 150 m) in Dubai. *Wikipedia*.

Name	City	Height (m)
Shanghai Tower	Shanghai, China	632
Burj Khalifa	Dubai, UAE	828
Marina 101	Dubai, UAE	425
Shangri-La Hotel	Dubai, UAE	200

Appendix: Derivation of the robust counterpart

We can simply remove the minimization. The constraint then becomes

$$\overline{a_1}x_1 + \overline{a_2}x_2 + \overline{a_3}x_3 + \overline{a_4}x_4 - (\Gamma z + q_1 + q_2 + q_3 + q_4) \geq b.$$

Thus, the production problem under budgeted uncertainty (robust counterpart) is given by

$$\begin{aligned}
\min_{\boldsymbol{x},z,\boldsymbol{q}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{s.t.} \quad & \overline{\boldsymbol{a}}^\top \boldsymbol{x} - \Gamma z - \sum_{i=1}^{4} q_i \geq b \\
& z + q_i \geq d_i x_i \quad i = 1, 2, 3, 4 \\
& \boldsymbol{x}, \boldsymbol{q} \geq \boldsymbol{0} \\
& z \geq 0.
\end{aligned}$$

Appendix: Less math via vertex enumeration

Consider $\Gamma = 1$. The vertices of Ξ^{budget} are $(0,0,0,0)$, $(1,0,0,0)$, ..., and $(0,0,0,1)$ (five in total).

In our specific setting, we only need to consider the last four. Let Ξ^{vertices} be the collection. The robust LP is

$$\begin{aligned} \min_{\boldsymbol{x}} \ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t. } & \bar{\boldsymbol{a}}^\top \boldsymbol{x} - \sum_{i=1}^{4} d_i \xi_i x_i \geq b \quad \forall \boldsymbol{\xi} \in \Xi^{\text{vertices}} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

The optimal solution is feasible for all $\boldsymbol{\xi} \in \Xi^{\text{budget}}$. In general, if the uncertainty set is a *polyhedron* (intersection of half spaces), we can perform vertex enumeration. Duality is not required, but we may need some basic binomial counting and polyhedral knowledge (so perhaps more math).

Appendix: Less math via vertex enumeration

```
v2 = np.array([1, 0, 0, 0])
v3 = np.array([0, 1, 0, 0])
v4 = np.array([0, 0, 1, 0])
v5 = np.array([0, 0, 0, 1])
model.addConstr(a @ x >= b)
model.addConstr(a @ x - v2 * d * x >= b)
model.addConstr(a @ x - v3 * d * x >= b)
model.addConstr(a @ x - v4 * d * x >= b)
model.addConstr(a @ x - v5 * d * x >= b)
```